AI-Driven Closed-Loop Precision Spraying for Real-Time Application Optimization

ABE 598 Digital Agriculture · Midterm Project Report · due October 21, 2026
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Draft built October 18, 2026. Items marked **TEAM TO CONFIRM** need a team member's check before submission.

1 Summary

All planned modules are built, tested and merged: simulator, nozzle catalog (316 chart rows, 547 candidate settings), predictive model, selector, scenario library, water-sensitive-paper (WSP) renderer, classical (B0) and learned (B1) verifiers, feedback module, loop harness, integration, experiment grid and a Streamlit demo. Five of the seven criteria in Table 1 of the proposal are met inside our simulator: the simulator, the predictive model (coverage MAE 2.84 pp random and 2.80 to 2.98 pp on withheld regimes, ridge 3.32 pp), the selector (97.0% requirement-met, with lower drift and about half the volume of the matched fixed setting), synthetic-card verification (B0 MAE 1.44 pp; B1 0.42 pp from a 6-epoch CPU checkpoint) and end-to-end operation. **The closed-loop criterion is missed:** the fast loop recovers 58% [38, 69] of the open-loop-to-oracle gap but uses +15.2% [+13.7, +16.9] more volume than the oracle (limit 10%), mostly because the oracle sees the per-card spread factor, and the slow loop as implemented did not help (-6% [-44, 17] recovery). Still open are the full GPU training of B1 and the real-card criterion, which is untested because the annotated cards are restricted. All results come from our own simulator and renderer, not the field. Milestone M1 was 16 days late and M2 2 days late, but all modules were built and merged on Oct 18; the remaining M3 work is B1 GPU training, real-card evaluation and a slow-loop redesign.

2 Technical Progress and Early Results

2.1 System architecture as built

The loop follows the proposal (Figure 1). Modules exchange frozen, versioned records defined in `contracts.py` (schema 1.0), so each was built and tested against mocks first. `loop.py` runs a 40-segment pass under five policies (`open`, `fast`, `fast_slow`, `oracle`, `best_fixed`) and logs one row per segment. The code base is 6,125 lines of Python with 180 test functions.

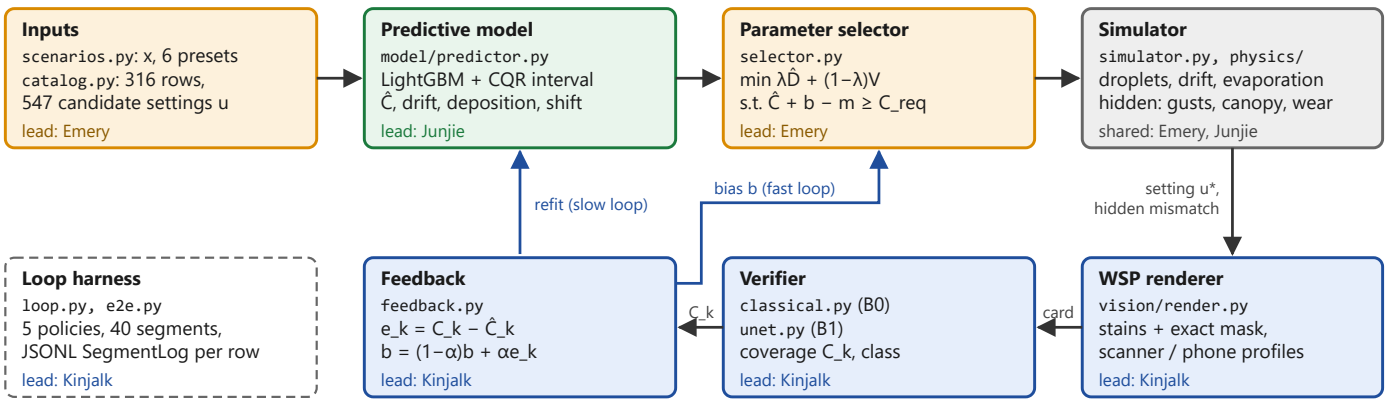


Figure 1. CoverageVision loop as implemented, with source modules. Border color marks the module lead (orange Emery, green Junjie, blue Kinjalk, grey shared). Black arrows: forward pass for segment k; blue arrows: fast loop (bias b to the selector) and slow loop (model refit between passes).

2.2 Nozzle catalog

`data/catalog/nozzles.csv` holds 316 chart rows for the five families (TP65 50, XR 72, DGX40 24, ULD 80, 3D 90; 43 family-size combinations). `catalog.py` interpolates flow along $Q \propto \sqrt{P}$, recomputes GPA at 20 in spacing, and generates 547 candidate settings on a 5 psi grid within each family's operating range. The validator reports 0 violations on four checks: flow within ±5% of the square-root law between adjacent rows (3 further deviations excused as chart rounding), GPA against 8 sampled chart cells, droplet class monotone in pressure, and ranges against `spec.yaml`. Compromises: TP65 droplet class uses the 80° column as proxy (D3), DGX40 contributes flow only (D4), ULD and 3D rows come from the manufacturers' sheets [12], and spray angles are nominal (D5).

2.3 Simulator physics

The simulator composes first-order relationships and is not a validated drift model. Flow follows $Q = Q_{\mathrm{ref}}\sqrt{(P/P_{\mathrm{ref}})}$ and application rate follows GPA = 5940·GPM/(MPH·W) [4]. The volume spectrum is log-normal with VMD set by the ASABE S572.3 class [3] (centers F 145, M 215, C 310 µm; relative span 1.2). Terminal velocity V_t uses a Schiller–Naumann drag correlation [9]. Evaporation follows the d^2 law with $\lambda = 84.76{\cdot}\Delta T_{\mathrm{wb}}$ µm^2/s [10] and Stull's wet-bulb approximation [11]; fully evaporated volume is lost. Droplets leave the nozzle at $v_0 = 0.6\sqrt{(2P/\rho_w)}$ (14.1 m/s at 40 psi) and relax toward V_t and the wind under linear drag ($\tau = V_t/g$); size bins displaced over 0.5 m cross-track are drift. Coverage uses a Boolean (Poisson) model [5],

$$C = 1 - \exp(-N{\cdot}\mathrm{E}[\pi s^2/4]/A), \quad s = \mathrm{SF}{\cdot}d, \quad \mathrm{SF} \sim \mathrm{U}[1.6, 2.2] \text{ per segment [6]}, \quad d \geq 50\ \mu\mathrm{m}.$$

Hidden mismatch (AR(1) gusts, canopy interception $\eta \in [0.6, 1.0]$, nozzle wear) is set by one level (none, low, med, high) and never seen by the model. Tests cover the anchors in Table 1, monotone trends (Figure 2), the Boolean formula against Monte Carlo placement, and seeded determinism.

Table 1. Physical anchors checked by tests (simulator values computed from the code at build time; drift anchors at 15 GPA, 2 m/s crosswind, H = 0.5 m, 25 °C, 50% RH, 40 psi, mean of 6 seeds). Plausibility checks, not field validation.

Quantity	Reference or accepted band	Simulator
Terminal velocity, 20 µm droplet (m/s)	0.012 ± 25%	0.012
Terminal velocity, 100 µm droplet (m/s)	0.265 ± 25%	0.248
Terminal velocity, 200 µm droplet (m/s)	0.7 ± 25%	0.707
Terminal velocity, 400 µm droplet (m/s)	1.8 ± 25%	1.610
Drift fraction, fine spray (F), 2 m/s crosswind	10% to 35%	30%
Drift fraction, medium spray (M), 2 m/s crosswind	2% to 15%	8%
Drift fraction, coarse spray (C), 2 m/s crosswind	below 5%	1%

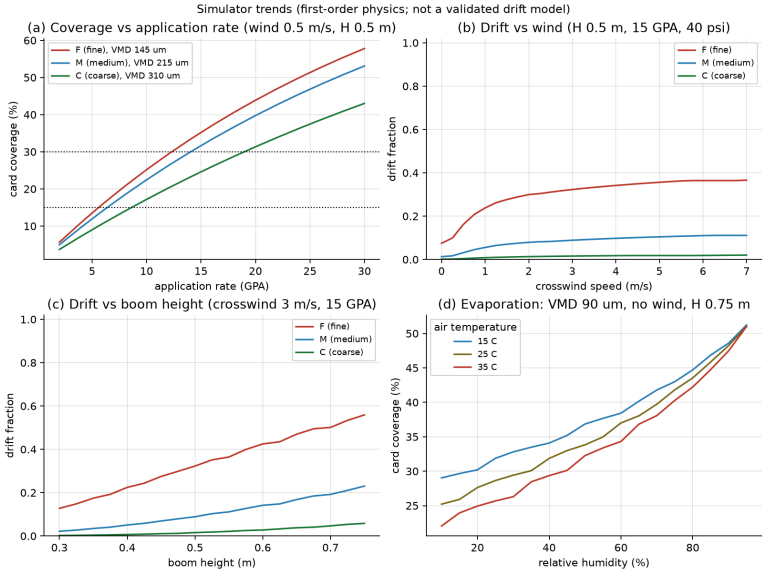


Figure 2. Simulator trends (first-order physics, not field-validated). (a) Coverage rises with application rate and falls for coarser classes at equal volume (dotted: 30% and 15% class limits). (b) Drift versus crosswind flattens above about 2 m/s. (c) Drift rises with boom height. (d) For a 90 µm spray, coverage falls with temperature and rises with humidity.

2.4 Dataset and predictive model

Dataset v1 has 50,000 simulator rows: Latin-hypercube conditions over the full operating ranges, settings drawn uniformly from the 547 candidates, no hidden mismatch. Coverage has median 17.3% (10th to 90th percentile 5.4% to 46.3%); 43.5% of rows are critical, 30.9% marginal, 25.6% acceptable. Features are raw conditions, wind as cross-track and along-track components, catalog-derived flow, VMD and angle, and physics features (wet-bulb depression, H/V_t, $|U_{cross}|H/V_t$, GPA/VMD). We compare ridge, random forest and LightGBM [15]; the interval comes from quantile LightGBM with conformal calibration [16]. Splits are a random 80/20 hold-out and three grouped splits withholding wind above 5 m/s, the whole 3D family, and the hot-dry regime (T > 35 °C, RH < 30%). LightGBM reaches a coverage MAE of 2.84 pp on the random hold-out (target 3 pp) against 3.32 pp for ridge and 3.23 pp for random forest, and is the best model on every split. Its 80% prediction interval covers 80.0% of held-out rows with mean width 9.0 pp. Scoring all 547 candidates takes 9.3 ms (median) and a 50,000-row refit takes 2.3 s, so the selector can run for every segment.

Table 2. Coverage MAE (pp) on held-out data against sampled outcomes; bold marks the best model. "vs expected" is LightGBM's MAE against noise-free expected coverage. Target: MAE ≤ 3 pp and below ridge.

Split	Ridge	Random forest	LightGBM	LightGBM vs expected	Drift MAE (fraction)	Shift MAE (mm)
Random 80/20	3.32	3.23	**2.84**	0.64	0.0030	3.4
Wind > 5 m/s withheld	3.41	3.38	**2.83**	0.84	0.0051	8.3
3D family withheld	3.42	3.57	**2.98**	0.90	0.0072	12.2
Hot-dry withheld	3.16	3.17	**2.80**	0.67	0.0034	3.6

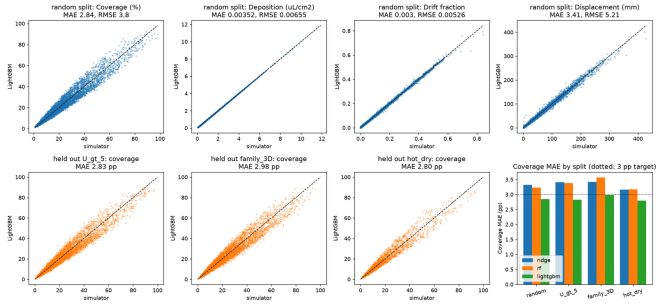


Figure 3. LightGBM versus simulator on held-out rows: four outputs on the random split (top), coverage on the withheld regimes (bottom left three) and coverage MAE by split and model (bottom right; dotted: 3 pp target).

2.5 Selector and scenarios

The selector implements Eq. (2) of the proposal: it scores every catalog candidate with the learned model (`SprayModel`, Section 2.4), keeps those with $C + b - m \geq C_{req}$, and picks the one minimizing $\lambda \cdot \text{drift}_n + (1 - \lambda) \cdot \text{volume}_n$ (min-max normalized over the feasible set; λ = 0.6). The margin m is the lower half-width of the prediction interval, floored at 1.5 pp. If nothing is feasible it returns the highest-coverage

setting and flags the segment. `best_fixed` is the single setting with the highest success rate over the scenario set, ties broken by lowest drift.

Table 3. Selector against fixed settings on 200 random no-mismatch scenarios at C_{req} = 30% (success = share with true coverage ≥ C_{req}; target ≥ 90% at lower drift than the matched fixed setting). In the first row the learned model selects and the simulator scores; all 547 candidates are scored per call.

Policy	Success (%)	Mean drift (fraction)	Mean volume (GPA)	Time per selection (ms)
Selector, learned model (config `spray`)	**97.0**	**0.0131**	**24.8**	**19.8**
Lowest-drift fixed setting with matching success: XR 15 @ 45 psi	98.0	0.0167	47.5	–
Best fixed setting overall: TP65 15 @ 55 psi	100.0	0.0171	52.5	–
Selector, simulator as model (config `oracle`)	38.5	0.0106	17.3	557.7

With the learned model the selector meets the requirement in 97.0% of the 200 scenarios (target ≥ 90%). The lowest-drift fixed setting that matches this success (XR 15 @ 45 psi, 98%) has mean drift 0.0167 against 0.0131 (21% lower) and applies 47.5 GPA against 24.8, so the selector uses about half the volume. The overall best fixed setting (TP65 15 @ 55 psi) reaches 100% but at drift 0.0171 and 52.5 GPA. One selection takes 20 ms including 17 ms of model time, so it can run for every segment. Raising λ from 0 to 1 moves the selector from 0.105 drift at 17.9 GPA to 0.0072 drift at 33.6 GPA, with success at or above 93.5% throughout (Figure 4).

Why the interval-based margin is needed. In config `oracle` the simulator both selects and scores (an expected-value model with no prediction interval), yet success is only 38.5%. Without an interval the selector keeps only the fixed 1.5 pp margin floor, and the per-segment random spread factor of the card pushes about half of the just-feasible choices below C_{req}. Its low drift (0.0106) and volume (17.3 GPA) are therefore bought with failures, not a better selector. The learned model's conformal interval supplies the margin that this noise requires, which is why the margin is taken from the interval and not set to a constant.

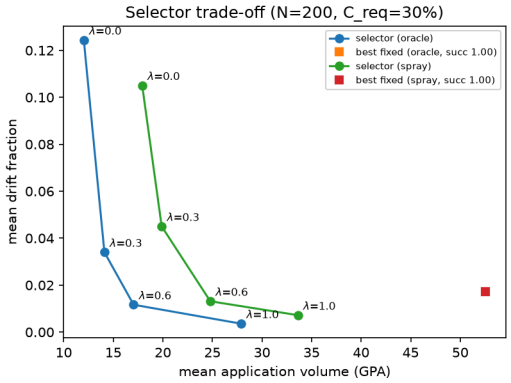


Figure 4. Mean drift against mean volume as λ varies (0, 0.3, 0.6, 1.0). Green: learned model selects (`spray`); blue: simulator selects and scores (`oracle`); square: best fixed setting (100% success under both). The `spray` curve lies to the right because the interval margin costs volume; that margin is what keeps its success high.

The scenario library generates 40-segment passes with AR(1) wind and gust bursts, rotating wind direction, diurnal temperature and humidity trends, and back-and-forth headings (the crosswind changes sign every 10 segments). Six presets each have an agronomic rationale: `calm_morning` (1.2 m/s base wind, benign reference), `gusty_afternoon` (4.0 m/s, gust probability 0.12), `hot_dry` (33 to 38 °C, RH 28% to 16%; hot, dry air speeds evaporation and raises drift potential [13]), `crosswind_shift` (wind direction rotates 140°), `marginal_limits` (6.0 m/s mean, near the label limit [14], exercises the infeasible path) and `inversion_evening` (0.6 m/s, cooling; a limitation test, since inversions are not modeled). The parameters are plausible regime choices, not measured weather statistics. The rationale text was drafted from Emery's specification and awaits his review **TEAM TO CONFIRM**.

2.6 WSP renderer and public data

The renderer turns the droplets on a card into an RGB image and an exact mask (76 × 26 mm, 600 dpi). Each detectable droplet leaves an area-preserving elliptical stain of diameter SF·d; a vectorized rasterizer replaces `cv2.ellipse`, which misses 10 to 15% of small stains' area. Degradations (warp applied to the mask too, illumination, blur, noise, JPEG) come in `scanner` and `phone` profiles (Figure 5). Over 100 undegraded cards at 1% to 80% coverage, mask coverage matches the analytic Boolean value to a mean absolute 0.14 pp (maximum 0.70 pp; test limit 2 pp), and worst-case render time is under 1 s per card.

The Simões et al. record 10.5281/zenodo.13995950 [7] provides 300 synthetic cards with polygon labels (CC BY 4.0; 635,839 droplets; coverage 1.5% to 41.0%, mean 13.0%). `vision/real_data.py` rasterizes the polygons and tiles cards; pixel size is inferred from card width because no dpi is published. The two annotated real cards (2,754 droplets) are in a later version whose files are restricted (D9). Access must be requested through Zenodo **TEAM TO CONFIRM once sent**; an email to the authors is drafted. Until access is granted the real-card criterion is untested.

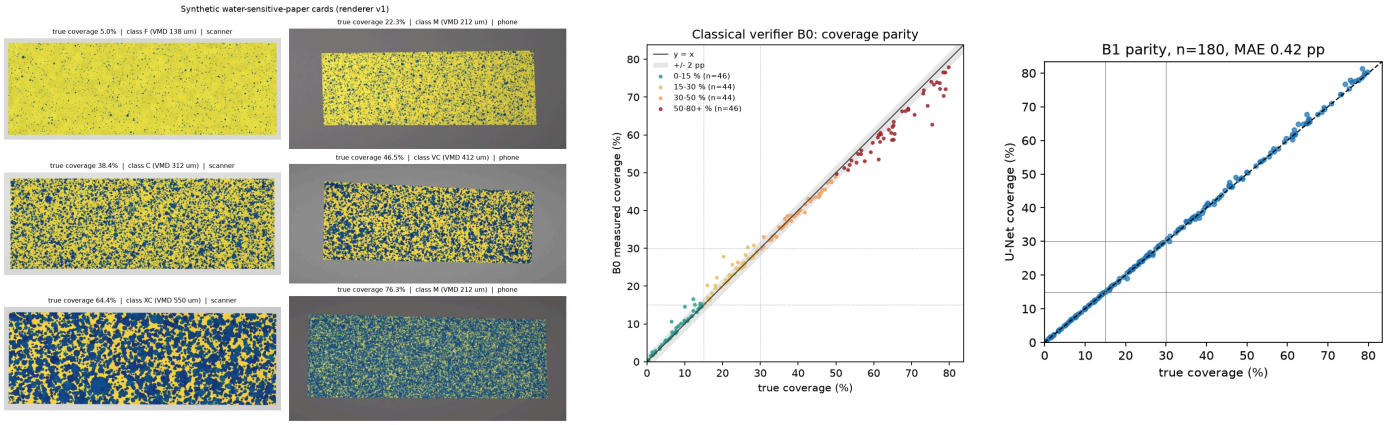


Figure 5. Left: synthetic cards from the team's renderer, 5% to 76% true coverage, scanner and phone profiles. Middle: classical verifier B0 coverage versus true coverage on the held-out synthetic test split (shaded: ±2 pp). Right: B1 (6-epoch CPU checkpoint, Section 2.7) on the same split; lines mark the 15% and 30% class limits.

2.7 Post-spray verification

Baseline B0 is a classical pipeline: card detection with perspective correction, illumination flattening, an (R−B) stain channel thresholded with Otsu's method [2,17], 3×3 opening, and a distance-transform watershed for stain counts and Dv0.1/0.5/0.9. A card is flagged saturated above 60% coverage, or above 30% when one connected component holds over 70% of the stained area.

On the 180 held-out synthetic test cards from the team's renderer (Table 4), B0 reaches a coverage MAE of 1.44 pp (target ≤ 2 pp), three-class agreement of 96.1% (target ≥ 90%) and a runtime of 0.29 s mean and 0.93 s maximum per card on a laptop CPU (target ≤ 2 s). Error is not uniform: in the 50-80+% band the MAE is 3.28 pp with a bias of -3.28 pp (67% of those cards are flagged saturated) and the worst single card is off by 12.8 pp. B0 underestimates coverage above 50% because stains merge. Stain counts and Dv values are unreliable (median relative count error 0.64 overall, 0.25 in the lowest band), which is why coverage, not counts, is the quantity the loop uses.

Table 4. Verifier accuracy on the 180 synthetic test cards by true-coverage band (B0 and B1 use different band edges): coverage MAE and signed bias in pp, three-class agreement, share of cards flagged saturated, mean stain-mask IoU and median relative stain-count error (– = not computed for B1). B1 rows are a 6-epoch CPU checkpoint, not the final model.

Verifier	True coverage	n	MAE (pp)	Bias (pp)	Class agreement (%)	Saturated (%)	IoU	Count error
B0	0-15%	46	0.70	+0.69	91.3	0	0.80	0.25
B0	15-30%	44	0.97	+0.86	95.5	0	0.82	0.50
B0	30-50%	44	0.77	-0.47	97.7	0	0.83	0.71
B0	50-80+%	46	3.28	-3.28	100.0	67	0.86	0.92
B0	**all**	**180**	**1.44**	**-0.57**	**96.1**	**17**	**0.83**	**0.64**
B1 (6-epoch CPU checkpoint)	0-10%	29	0.11	+0.10	100.0	–	0.91	–
B1 (6-epoch CPU checkpoint)	10-20%	27	0.18	+0.12	96.3	–	0.92	–
B1 (6-epoch CPU checkpoint)	20-30%	34	0.38	+0.25	100.0	–	0.92	–
B1 (6-epoch CPU checkpoint)	30-45%	34	0.45	+0.23	100.0	–	0.93	–
B1 (6-epoch CPU checkpoint)	45-60%	22	0.54	+0.49	100.0	–	0.93	–
B1 (6-epoch CPU checkpoint)	60-100%	34	0.82	+0.75	100.0	–	0.95	–
B1 (6-epoch CPU checkpoint)	**all**	**180**	**0.42**	**+0.33**	**99.4**	**–**	**0.93**	**–**

B1 (6-epoch CPU checkpoint). B1 is a U-Net with a lightweight pretrained encoder [8] (`vision/unet.py`); the weights evaluated here come from a **6-epoch run on a laptop CPU** and understate the final model. On the same 180 test cards the checkpoint reaches a coverage MAE of 0.42 pp (bias +0.33 pp), stain-mask IoU 0.93 (B0: 0.83) and three-class agreement of 99.4% (B0: 1.44 pp MAE). Error grows with coverage (0.11 pp in the 0-10% band, 0.82 pp in the 60-100% band). **Runtime:** on an idle laptop CPU over 16 test cards the first call took 1.74 s (including warm-up) and later calls averaged 1.20 s per card (maximum 1.48 s), inside the 2 s target; an earlier timing over all 180 cards while the CPU was busy with the experiment grid (mean 1.90 s, maximum 7.4 s) is not used for the criterion. Like B0, B1 is tested only on cards from the same generator (Figure 5, right). **Next:** full GPU training (NCSA H100, or the Colab notebook `notebooks/train_unet_gpu.ipynb`), then re-evaluation and re-timing on all 180 cards.

2.8 Closed-loop experiments

The grid (`experiments/closed_loop.py`) runs five policies over four presets (`calm_morning`, `crosswind_shift`, `gusty_afternoon`, `hot_dry`), four hidden-mismatch levels (none, low, med, high) and 20 seeds, 40 segments per pass at C_{req} = 30%, all inside the simulator. The headline model is the learned `SprayModel`; a run with the mismatch-free simulator physics as the model (`oracle_nominal`) gives the same picture (fast loop 60% [43, 71] recovery, +13.8% volume). The verifier is *noisy*: perfect coverage plus Gaussian error with per-band bias and standard deviation from the B0 results. Gap recovery is (closed − open)/(oracle − open) in the share of segments meeting C_{req}; volume overhead is mean applied GPA relative to the oracle. Headline numbers pool low, med and high mismatch; brackets are 95% paired-bootstrap CIs over seeds.

Table 5. Closed-loop success criterion (proposal Table 1), learned model.

Proposal criterion (Table 1, "Closed loop")	Policy	Measured (95% CI)	Verdict
Recovers ≥ 50% of the open-to-oracle gap in requirement-met segments	fast loop	58% [38, 69]	met on the point estimate; the CI includes values below 50%
	fast + slow loop	−6% [−44, 17]	missed
≤ 10% more volume than oracle	fast loop	+15.2% [+13.7, +16.9]	missed
	fast + slow loop	+6.2% [+5.2, +7.2]	met
Both conditions together	none of the policies	fast loop meets recovery but not volume; fast + slow meets volume but not recovery	**MISSED**

The closed-loop criterion is missed. The fast loop recovers 58% [38, 69] of the open-loop-to-oracle gap, which meets 50% on the point estimate but not with confidence, and uses +15.2% [+13.7, +16.9] more volume than the oracle against the 10% limit. Adding the slow loop meets the volume limit (+6.2% [+5.2, +7.2]) but not recovery (−6% [−44, 17]).

The fast loop works as designed. The requirement-met share under mismatch stays at 0.94 (low 0.947, med 0.943, high 0.932; Table 6, Figure 6), while open loop falls from 0.954 (low) to 0.864 (med) and 0.762 (high) and the best fixed setting to 0.61 at high. Recovery by preset is 69% (`calm_morning`), 54% (`crosswind_shift`), 54% (`gusty_afternoon`) and 48% (`hot_dry`, CI [20, 66]). At low mismatch the open-to-oracle gap is only 4.6 pp (the model's margin already protects open loop), so recovery there is indistinguishable from zero; the criterion is informative at med and high mismatch (gaps 13.6 and 23.8 pp).

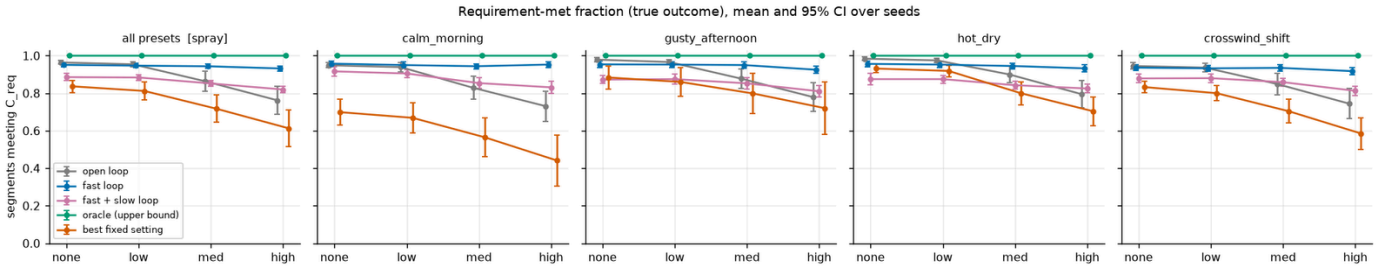


Figure 6. Requirement-met share of segments (true outcome) against hidden-mismatch level, learned model: all presets pooled (left) and per preset; mean and 95% CI over 20 seeds.

Table 6. Share of segments meeting C_{req} by policy and mismatch level (learned model; mean ± 95% CI over 20 seeds; presets pooled; the oracle row, 1.000 at every level, is omitted).

policy	none	low	med	high
open	0.964 ± 0.010	0.954 ± 0.015	0.864 ± 0.053	0.762 ± 0.076
fast	0.951 ± 0.012	0.947 ± 0.011	0.943 ± 0.012	0.932 ± 0.013
fast_slow	0.886 ± 0.019	0.884 ± 0.015	0.852 ± 0.017	0.820 ± 0.019
best_fixed	0.837 ± 0.032	0.812 ± 0.048	0.717 ± 0.072	0.613 ± 0.098

Why the volume criterion is missed. With no mismatch at all the fast loop already uses +12.2% [11.2, 13.1] more volume than the oracle. The oracle sees the per-segment random spread factor of the card (irreducible noise, standard deviation about 5 pp of coverage) and needs no safety margin, whereas every model-based policy must keep a margin from its prediction interval. Only about 3 pp of the 15.2% comes from correcting the hidden mismatch. The same noise bounds how much of the residual a bias correction can remove: raw residual variance stays at about 27 to 29 pp^2 inside a pass. The 10% limit therefore compares against a reference no deployable controller can match (Section 4.1).

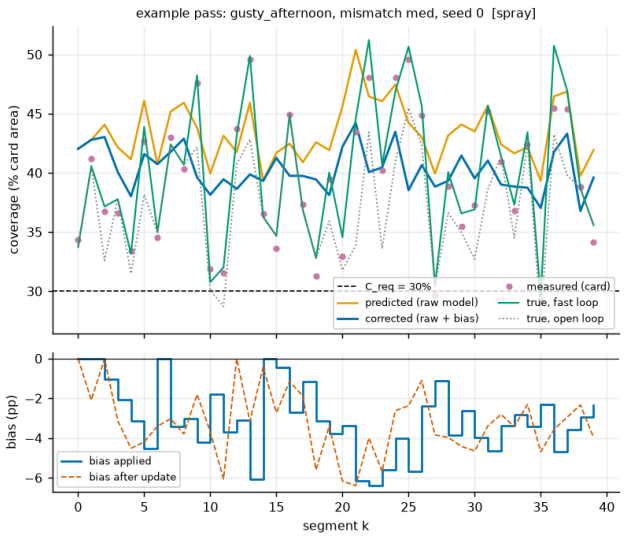


Figure 7. One pass (`gusty_afternoon`, med mismatch, seed 0, learned model). Top: raw and bias-corrected prediction, measured card values, and true coverage under the fast loop and open loop. Bottom: bias applied to each segment and bias after the update (0 to about −6 pp).

The slow loop, as implemented, did not help. It refits `SprayModel` on the measured cards of earlier passes with record weight 30. It removes the average model error (mean raw residual at high mismatch −6.0 pp in pass 1, −1.8 pp in pass 3) and cuts volume, but requirement-met drops to 0.85 against 0.94 for the fast loop and the raw residual variance rises from about 27 to about 33 pp^2. A 6-seed diagnostic with no mismatch (`experiments/slow_loop_diagnostic.py`) shows the 80% prediction interval narrowing after the refit (mean half-width 8.0 to 5.2 pp), true coverage falling below its lower bound in 21 to 27% of segments instead of 7.5%, and the MAE rising from 4.7 to 5.0–5.1 pp. A plausible cause, not a tested mechanism, is that a few dozen noisy, heavily weighted records are fitted too closely and the conformal calibration is not redone on them. Record weights 1, 5 and 30 give 36%, 10% and −20% recovery (Table 7); 30, the grid's value, is the worst. The slow loop needs a better update rule (Section 6).

Ablations (fast loop, `gusty_afternoon`, med mismatch, 20 seeds unless noted). A smaller α is cheaper (α = 0.1, 0.3, 0.6 give 45%, 59%, 54% recovery at +13.0%, +17.9%, +20.5% volume) at a small loss of reliability. A global bias recovers more (68%) than per-regime biases (59%) at +19.8% against +17.9% volume, plausibly because each of four regimes sees few cards before its warm-up gate opens; warm-up of 0, 2 or 4 cards (57%, 59%, 50%) differs by less than the CI. Perfect, noisy and rendered-B0 verifiers give 51%, 59% and 44% (B0: 5 seeds, CI [−200, 90]), inside the seed-to-seed spread, so measurement error is not what limits the loop. Retreat flags were never raised (0 in 1,600 learned-model passes).

Table 7. Ablations of the fast loop and of the slow-loop record weight (learned model; warm-up and the repeated baseline row omitted).

group	variant	seeds	met	gap recovery	volume vs oracle
alpha	alpha=0.1	20	0.932 ± 0.024	45% [28, 60]	+13.0% [+12.0, +13.9]
alpha	alpha=0.3	20	0.950 ± 0.016	59% [33, 74]	+17.9% [+16.5, +19.3]
alpha	alpha=0.6	20	0.944 ± 0.013	54% [27, 69]	+20.5% [+18.2, +23.1]
bias scope	per-regime bias	20	0.950 ± 0.016	59% [33, 74]	+17.9% [+16.5, +19.3]
bias scope	global bias	20	0.961 ± 0.013	68% [48, 80]	+19.8% [+17.9, +22.1]
verifier	verifier=perfect	20	0.940 ± 0.015	51% [25, 66]	+15.2% [+14.1, +16.4]
slow-loop record weight	slow loop, record weight 1 (3 passes, final pass)	20	0.921 ± 0.021	36% [−2, 58]	+14.2% [+12.5, +15.9]
slow-loop record weight	slow loop, record weight 5 (3 passes, final pass)	20	0.890 ± 0.027	10% [−39, 39]	+11.9% [+10.3, +13.4]
slow-loop record weight	slow loop, record weight 30 (3 passes, final pass)	20	0.853 ± 0.030	−20% [−83, 15]	+6.9% [+5.5, +8.1]
verifier	verifier=b0 (rendered, 5 seeds)	5	0.975 ± 0.031	44% [−200, 90]	+14.6% [+12.1, +17.0]

What these runs do and do not show. Plant, model training data, renderer and hidden mismatch all come from one hand-built simulator, so the runs test whether the *control logic* behaves as designed when the model is wrong in a known, low-dimensional way; they do not measure field efficacy. We injected the mismatch (gust multiplier, canopy interception, nozzle wear), and it is of the kind the bias term is built to absorb; inversions, boom roll, clogging and card-placement error are not tested, and the default verifier omits heavy-tailed failures and real-card domain shift.

2.9 Demonstration

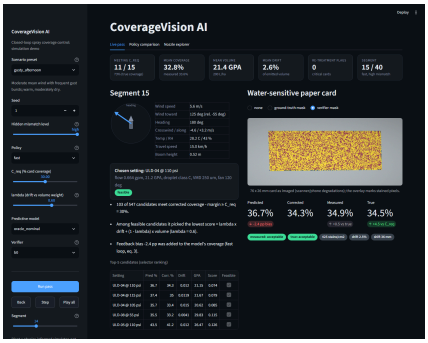


Figure 8. Demo app, Live pass tab (top of the page): scenario `gusty_afternoon`, high hidden mismatch, fast policy, segment 15 of 40, with chosen setting, selector reasoning, rendered card and the four coverage values.

The Streamlit app `app/main.py` runs a 40-segment pass segment by segment and shows every decision: the conditions, the chosen setting and why (feasible candidates among the 547, top-5 ranking), the rendered card with the verifier's mask, and the predicted, corrected, measured and true coverage. It has three tabs. **Live pass** steps through one pass under a chosen preset, mismatch level, policy, model and verifier. **Policy comparison** runs chosen policies (default open, fast and oracle) on the same seed and hidden mismatch. **Nozzle explorer** shows catalog-derived quantities and the nominal simulator's response for a chosen nozzle.

3 Status Against the Success Criteria

Table 8 mirrors Table 1 of the proposal. **met**: target satisfied on the evaluation available now (inside our simulator only); **missed**: evaluated and the target was not reached (causes in Section 2.8); **at risk**: blocked by an outside dependency; **pending**: no result yet. Simulator and model are unchanged since the first draft.

Table 8. Success criteria (proposal Table 1) and current status.

Criterion	Target	Current measured	Status
End-to-end	A simulated pass runs from prediction to correction with no manual steps; every decision is logged and visualized.	`uv run python -m coverageai.e2e --real` runs a 40-segment pass with rendered cards and B0 in 33 s (mean |measured − true| 0.57 pp). The Streamlit demo (`app/main.py`) logs and visualizes every decision (Section 2.9).	**met**
Simulator	Coverage rises with volume; drift rises with wind, release height and fine fraction.	36 simulator tests pass, including monotone sweeps for each trend and the D10 anchors (Table 1).	**met**
Predictive model	Coverage MAE ≤ 3 pp on held-out conditions and below the linear baseline; grouped-regime error reported; sensitivity checks pass.	2.84 pp random; 2.83, 2.98, 2.80 pp on three withheld regimes; ridge 3.32 pp; 5 of 5 sensitivity checks pass. Noise floor 2.76 pp.	**met**
Selector	No mismatch: ≥ 90% of scenarios meet C_{req} at lower drift than the best fixed setting with the same success.	Learned model: success 97.0%, drift 0.0131, 24.8 GPA, against 98%, 0.0167, 47.5 GPA for the matched fixed setting: lower drift, about half the volume (Table 3).	**met**
Verification, synthetic cards	MAE ≤ 2 pp; three-class agreement ≥ 90%; ≤ 2 s per card on a laptop CPU.	B0: MAE 1.44 pp, agreement 96.1%, 0.93 s worst case (180 cards). B1 (6-epoch CPU checkpoint): MAE 0.42 pp, agreement 99.4%, 1.20 s mean and 1.48 s maximum per card on an idle CPU (16 cards). Full GPU training of B1 pending.	**met**
Verification, real cards	MAE ≤ 5 pp on real cards.	Untested. The two annotated real cards are in a restricted Zenodo version (D9); access must still be requested through Zenodo **TEAM TO CONFIRM**.	**at risk**
Closed loop	Under mismatch, recovers ≥ 50% of the open-loop-to-oracle gap in requirement-met segments with ≤ 10% more volume than the oracle (20 seeds, 95% CI).	Fast loop: recovery 58% [38, 69] (meets 50% on the point estimate, CI crosses it) but volume +15.2% [+13.7, +16.9] against the oracle (limit 10%). Fast + slow loop: recovery -6% [-44, 17], volume +6.2% [+5.2, +7.2]. Causes: the oracle sees the per-card spread factor (+12.2% volume even with no mismatch), and the slow loop lowers reliability (Section 2.8).	**missed**

4 Analysis and Reasoning

4.1 What the results show and do not show

- **The model works, but the 3 pp target sits close to a floor set by the simulator.** Coverage depends on a per-row random spread factor, so no model can beat an MAE of 2.76 pp against sampled outcomes. LightGBM's 2.84 pp is 0.09 pp above that floor; against the noise-free expectation its MAE is 0.64 pp (ridge: 1.82 pp). Ridge's miss (3.32 pp) reflects real nonlinearity, but a 3 pp target leaves almost no slack, so the final report will also give error against expected coverage.
- **Withholding regimes barely changes coverage error but does change drift and shift.** Coverage MAE stays between 2.80 and 2.98 pp because coverage is governed by volume and droplet size, which training data span. Shift MAE grows from 3.4 mm to 8.3 mm (wind above 5 m/s withheld) and 12.2 mm (3D withheld); drift MAE from 0.0030 to 0.0051 (wind above 5 m/s), so drift estimates are least reliable at high wind, where they matter most.
- **Sensitivity sweeps pass** (5 of 5); they check direction, not magnitude.

- **B0 is accurate mid-range and degrades at high coverage.** B0 meets all three synthetic targets, but its error is 3.28 pp in the 50-80+% band against 0.70 pp in the lowest band (WSP saturation, as the proposal anticipated). Class thresholds sit at 15% and 30%, so this rarely changes the class; it would matter for a requirement above 50%.
- **The interval-based margin is what makes the selector reliable.** With the simulator as the model and only the 1.5 pp margin floor, the selector meets the requirement in 38.5% of scenarios; with the learned model and its conformal interval it does in 97.0%. The per-card spread factor is noise no model can predict, so the margin must come from a calibrated interval.
- **The closed-loop volume criterion compares against an oracle with information no deployable controller has.** The oracle sees each card's spread factor and needs no margin, which alone costs +12.2% volume with no mismatch; only about 3 of the 15.2 points are due to mismatch correction. A fairer oracle, one that knows the mismatch but not the per-card noise, is planned for M4.
- **Evaluation is circular.** Simulator, renderer, model and verifier share assumptions (spread-factor range, Boolean overlap, 50 µm cutoff), so good synthetic scores show consistency, not farm performance. The hidden mismatch, oracle policy and perfect-measurement ablation separate the loop's benefit from this circularity.

4.2 Key decisions and why

The decision log (`Docs/decisions.md`) holds ten entries, each resolving an ambiguity that would have split the modules. D1 and D2 fix conventions (wind direction is where the air moves toward; four distinctly named outputs, because "D" meant drift in the proposal and deposition in the specification). D3 to D5 record catalog gaps. D6 limits the selector to nozzle, size and pressure, deferring speed as a control variable. D7 and D8 record the compressed schedule and the use of system Python 3.14. D9 reports the real-card criterion as untested because the annotated cards are restricted, rather than estimating it.

D10, exit velocity. With the first-order displacement $x = U{\cdot}H/V_t$ and a 0.5 m threshold, a medium droplet (215 µm, V_t = 0.78 m/s) needs 0.64 s to fall 0.5 m and moves 1.3 m in a 2 m/s crosswind, so the model predicted 87% drift of medium-droplet volume, which is implausible at this height. Real droplets leave the nozzle at about 14 m/s and cross most of the height in tens of milliseconds before settling. Adding exit velocity with linear-drag relaxation gives 30% drift for fine, 8% for medium and 1% for coarse sprays at the same conditions. We kept the 0.5 m threshold instead of retuning it, so the fix changes the physics, not a free parameter.

4.3 Limitations

- **Same class, same physics.** Spray angle and pattern overlap are not modeled, so families with the same droplet class and flow are indistinguishable and the selector's choice among them is arbitrary (planned fix in Section 6). The spread factor also sets the 2.76 pp coverage floor, and real SF depends on paper, droplet size and humidity.
- **No real annotated cards yet**, so sim-to-real transfer of B0 and B1 is unmeasured; the proposal cites a drop from 57.06% to 40.81% for YOLOv8 from synthetic to real cards [7].
- **WSP physics limits.** Droplets below about 50 µm leave no stain and accuracy falls at high coverage, as B0's errors above 50% show.
- **Inversions and canopy structure are not modeled**; `inversion_evening` tests a limitation, and the drift constants are fitted to published trends, not measured.

5 Project Management and Teamwork

5.1 Schedule, tools and process

M1 and M2 passed without committed code, so the freeze, baselines and first integration were planned as three gates (Oct 18 to 20) before the midterm; in practice all modules were built and merged on Oct 18. Tools follow the proposal: a private GitHub repository with review before every merge, a Microsoft Teams channel, a decision log (10 entries) and an AI-use log (16 entries); tests and the end-to-end smoke test must pass before a merge.

AI-assisted workflow. One orchestrator session (Claude Opus 5.5) wrote the execution plan, froze the contracts and `spec.yaml`, and reviewed each work package. Implementation agents (Claude Sonnet models) each took one package in a separate git worktree against the frozen contracts, added tests and logged the session; packages ran in parallel and merged in dependency order. The accountable member reviews each module; contracts change only through the orchestrator; Kinjalk runs the orchestration and merges (Table 9).

Schedule. All Wave 1 to 3 modules (specification, catalog, simulator, model, selector, scenarios, renderer, B0, B1 code, feedback, harness, integration, experiments and demo) were built and merged on Oct 18. M1 (specification freeze, Oct 2) was met 16 days late, with contracts, `spec.yaml`, mocks and smoke test frozen. M2 (baselines, Oct 16) was met 2 days late, with simulator, dataset, model, scenarios, renderer and B0. The M3 work that remains (Oct 30) is B1 training on a GPU, real-card evaluation if access is granted, and the slow-loop redesign; the selector re-evaluation with the integrated model, also planned for M3, is already done (Section 2.5). M4 (Nov 20) and M5 (Dec 9: final report, demonstration, presentation) are unchanged, with the fairer-oracle comparison added to M4.

5.2 Individual contributions

"Work completed" means built, tested and merged unless stated. Modules implemented by AI agents are owned and reviewed by the named member; each member's own authoring and review is stated only where project documents show it.

Table 9. Contribution table.

Member	Responsibilities	Work completed	Next milestone
Emery Brewer (remote)	System specification (parameter definitions, physics relationships, assumptions); scenario rationale review; selector ownership; domain validation.	Authored *CoverageAIDraftParameters.pdf* (variables, units, ranges, physics relationships, validation approach) and the earlier simulations in *ProjectDeliveryProposal.pdf*; these define `spec.yaml` and the physics the simulator encodes. The catalog, scenario library with rationale text, and selector were implemented by AI agents for his ownership and are merged. **TEAM TO CONFIRM** his review of catalog values, preset rationale and selector, and any further contributions.	Review the selector evaluation and the closed-loop results (M3, Oct 30); pattern-overlap, spray-angle and speed-control relationships (M4, Nov 20); domain-validation section (M5, Dec 9).

Junjie Ao	Simulator and dataset pipeline; predictive model ownership.	The simulator (`physics/`, `simulator.py`, with the D10 fix), dataset v1 (50,000 rows) and predictive model (LightGBM with interval, ridge and forest baselines, grouped-split evaluation) are built, tested and merged; AI agents implemented them for his ownership. **TEAM TO CONFIRM** his own authoring, review and decisions on these modules.	Model v2 with error analysis, and interval recalibration for the slow-loop redesign (M3, Oct 30); model in the full loop (M4, Nov 20); evaluation of the spray-angle extension (M5).
Kinjalk Parth	WSP renderer; verification (B0, B1); feedback and loop harness; integration and experiments; orchestration of AI-assisted implementation.	Execution plan, contracts, `spec.yaml`, decision log, mocks, smoke test; renderer (0.14 pp mean agreement with the analytic model); B0 (1.44 pp MAE, 180 synthetic test cards); Simões loader; feedback module and loop harness (five policies); selector evaluation with the learned model (97.0% requirement-met); integration (`e2e --real`); closed-loop grid and ablations; Streamlit demo app (orchestrated); B1 code with partial 6-epoch CPU training (0.42 pp MAE); review and merge of all work packages; this report.	B1 trained on GPU and re-timed (M3, Oct 30); slow-loop redesign (M3); fairer oracle and extended closed-loop grid with B1 (M4, Nov 20); real-card evaluation if granted; final presentation (M5).

Risks update. The proposal's risks stand: sim-to-real transfer (unmeasured until real cards arrive), circular evaluation (hidden mismatch, oracle, perfect-measurement ablation) and integration slip (realized as the M1 and M2 slip). New: AI-written code that teammates must understand (owner review; each member presents their own module), a 3 pp model target near the noise floor, and the missed closed-loop criterion, addressed in M3 and M4 (Section 6).

6 Next Steps to M4 and M5

- **By Oct 30 (M3).** Train the U-Net B1 on a GPU (NCSA H100 or `notebooks/train_unet_gpu.ipynb` on Colab) and re-time it on all 180 test cards. Redesign the slow loop: a bias-only or shrinkage update, recalibration of the prediction interval after each refit, and record weight 1. Obtain the restricted real cards **TEAM TO CONFIRM** and evaluate B0 and B1 on real tiles, or on our own phone photos of WSP.
- **By Nov 20 (M4).** Add a fairer oracle that knows the mismatch but not the per-card spread factor, and rerun the closed-loop grid against both oracles with B1 and B0 as verifiers, all six presets and the redesigned slow loop (20 seeds, 95% CIs). Add spray angle and pattern overlap (Emery's relationships) so same-class families differ; make travel speed a control variable; polish the demo app.
- **By Dec 9 (M5).** Hardening, final report with every criterion in Table 8 measured, domain-validation section, and a presentation in which each member presents their own module.

7 AI-Use Disclosure

The course permits AI use, and this project uses it heavily. The execution plan, contracts, simulator, catalog transcription, model, selector, scenario library, renderer, classical and U-Net verifiers, feedback module, loop harness, integration, closed-loop experiments, demo app, tests and this report were produced by Claude models (Claude Opus 5.5 as orchestrator and reviewer, Claude Sonnet models as implementation agents) under the team's direction; 16 sessions are recorded in `Docs/ai_usage_log.md`. Team members set the scope, resolve ambiguities in the decision log, review the modules they own, and are responsible for every claim here. Numbers are read from results files at build time, except the end-to-end run time and the idle-CPU timing of B1, which the orchestrating session measured. Each member must verify the statements marked **TEAM TO CONFIRM** before submission.

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References [1]–[8] reuse the proposal's [1], [2], [8], [10], [11], [14], [9], [12]; the rest are sources used in the code or project documents.